
MS48-LR -- LoRaWAN To Modbus Gateway



SE



NO

Direktronik AB | Box 234 , 149 23 Nynäshamn | Besöksadress Konsul Johnsons väg 15 149 45 Nynäshamn
Telefon 08 52 400 700 | Fax | Epost info@direktronik.se | Org.nr 556281-9663 | Bankgiro 922-0179

Table of Contents

1. Introduction	5
1.1 What is MS48-LR	5
8. More Services	58



Table of Contents:

- [1. Introduction](#)
 - [1.1 What is MS48-LR](#)
 - [1.2 How does it work?](#)
 - [1.3 Specifications](#)
 - [1.4 Features](#)
 - [1.5 LED Indicators](#)
 - [1.6 RS485 & RS232 interface](#)
 - [1.7 Button Instruction](#)
 - [1.8 Installation](#)

- [2. Quick Start](#)
 - [2.1 Access and Configure MS48-LR](#)
 - [2.1.1 Find IP address of MS48-LR](#)
 - [Method 1: Connect via MS48-LR WiFi](#)
 - [Method 2: Connect via Ethernet with DHCP IP from the router](#)
 - [Method 3: Connect via MS48-LR Fallback IP](#)
 - [Method 4: Connect via WiFi with DHCP IP from the router](#)
 - [2.1.2 Access Configure Web UI](#)
 - [2.2 Typical Network Setup](#)
 - [2.2.1 Overview](#)
 - [2.2.2 Use the WAN port to access the Internet](#)
 - [2.2.3 Access the Internet as a WiFi Client](#)
 - [2.2.4 Use built-in 4G modem for internet access](#)
 - [2.2.5 Check Internet connection](#)
 - [2.3 Bridge LoRaWAN network to Modbus network](#)
 - [Step 1: Configure the LoRa Radio to your area Frequency Plan](#)
 - [Step 2: Copy the unique Gateway EUI & Configure the LoRaWAN Server address](#)
 - [Step 3: Enable the Built-in LoRaWAN Network Server](#)
 - [Step 4: Logging to the Built-in LoRaWAN Network Server](#)
 - [Step 5: Register the gateway to the built-in ChirpStack](#)
 - [Step 6: Register the Sensor-node to the built-in ChirpStack](#)
 - [1\) Add Device Profiles](#)
 - [2\) Add End Node Device](#)
 - [Step 7: Configure Modbus RTU/TCP Slave](#)
 - [1\) RTU Mode:](#)
 - [2\) TCP Mode:](#)
 - [3\) PLC\(Modbus server/master\) data show:](#)
 - [Step 8: Configure Sensor decode to Built-in Chirpstack](#)
 - [1\) Add Sensor's decode to Chirpstack](#)
 - [2\) Check the decode on ChirpStack](#)
 - [Step 9: Configure Modbus RTU/TCP Slave Sensor Decode](#)
 - [PLC\(Modbus server/master\) data show:](#)
 - [2.4 Accept data to registers for specified Fport](#)
 - [PLC\(Modbus server/master\) data show:](#)
- [3. Web Configure Pages](#)
 - [3.1 Home](#)
 - [3.2 LoRa Settings](#)
 - [3.2.1 LoRa --> LoRa](#)
 - [3.3 LoRaWAN Settings](#)
 - [3.3.1 LoRaWAN --> LoRaWAN Semtech UDP](#)
 - [3.3.2 LoRaWAN --> LoRaWAN Basic Station](#)
 - [3.4 Network Settings](#)
 - [3.4.1 Network --> WiFi](#)
 - [3.4.2 Network --> System Status](#)
 - [3.4.3 Network --> Network](#)
 - [3.4.4 Network --> Cellular](#)
 - [3.5 System](#)
 - [3.5.1 System --> System Overview](#)
 - [3.5.2 System --> System General](#)
 - [3.5.3 System --> Backup/Restore](#)
 - [3.5.4 System --> Remoteit](#)
 - [3.5.5 System --> Package Management](#)
- [4. Build-in Server](#)
 - [4.1 LoRaWAN Network Server -- ChirpStack](#)
 - [4.2 Application Server -- Node-Red](#)
 - [Using Node-Red, InfluxDB and Grafana](#)
 - [Upgrade the node.js](#)
- [5. Use RS232 Interface](#)
 - [5.1 Link Node-Red to Local ChirpStack](#)
- [6. Use RS485 Interface](#)

- [6.1 Initialize the GPIO21](#)
- [6.2 Set the RS485 Tx Mode](#)
- [6.3 Set the RS485 Rx Mode](#)
- [7. Power Analyze](#)
- [8. More Services](#)

1. Introduction

1.1 What is MS48-LR

The MS48-LR is an open-source LoRaWAN gateway that can **convert LoRaWAN to Modbus RTU/TCP**. It lets you bridge LoRa wireless network to an IP network via **WiFi , Ethernet or Cellular Network** (via Optional 4G module). The LoRa wireless allows users to send data and reach extremely long ranges at low data rates.

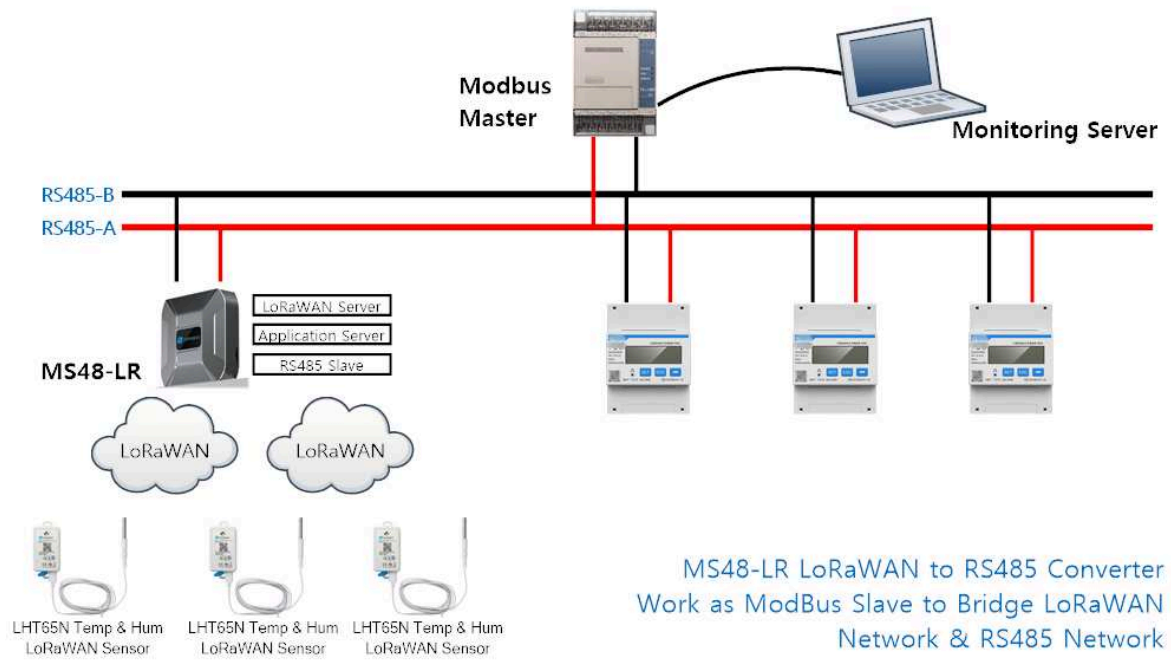
The MS48-LR is fully compatible with LoRaWAN protocol. MS48-LR includes a **built-in LoRaWAN Server and IoT server**, User can connect various LoRaWAN devices to MS48-LR, and use PLC to access these devices via Modbus network, MS48-LR eliminates the need to add additional lorawan gateways to implement **LoRaWAN to Modbus RTU/TCP**.

Different countries use different LoRaWAN frequency bands. MS48-LR has these bands pre-configured. Users can also customize the frequency bands to use in their own LoRa network.

MS48-LR supports **remote management**. System Integrator can easily remote monitor the gateway and maintain it.

1.2 How does it work?

The MS48-LR can run as a Modbus RS485-RTU slave, which converts the LoRaWAN sensor upstream data into modbus data and writes it into the register of the 03 function code



1.3 Specifications

Hardware System:

- CPU: Quad-core Cortex-A7 1.2Ghz
- RAM: 512MB
- eMMC: 4GB

Interface:

- 10M/100M RJ45 Ports x 1
- Multi-Channel LoRaWAN Wireless
- WiFi 802.11 b/g/n
- USB 2.0 host connector x 1
- Mini-PCI E connector x 1
- RS485 Interface x 1

- RS232 Interface x 1

LoRa Spec:

- Up to -140 dBm sensitivity with SX1250 Tx/Rx front-end
- 70 dB CW interferer rejection at 1 MHz offset
- Able to operate with negative SNR, CCR up to 9dB
- 8 x 8 channels LoRa packet detectors, 8 x SF5-SF12 LoRa demodulators, 8 x SF5-SF10 LoRa demodulators, 125/250/500 kHz LoRa demodulator and 1 x (G)FSK demodulator
- Dual digital TX & RX radio front-end interfaces
- 10 programmable parallel demodulation paths
- Dynamic data-rate (DDR) adaptation
- True antenna diversity or simultaneous dual-band operation

Cellular 4G LTE (optional):

- Quectel: [EC25 LTE module](#)
- Standard Size SIM Slot
- 2 x 4G Sticker Antenna.
- Up to 150Mbps downlink and 50Mbps uplink data rates
- Worldwide LTE, UMTS/HSPA+ and GSM/GPRS/EDGE coverage
- MIMO technology meets demands for data rate and link reliability in modern wireless communication systems

Operating Condition:

- Work Temperature: -20 ~ 70°C
- Storage Temperature: -20 ~ 70°C
- Power Input: 12V, 2A, DC

1.4 Features

- Open Source Debian system
- Managed by Web GUI, SSH via WAN or WiFi
- Remote Management
- Auto-provisioning for batch deployment and management
- LoRaWAN Gateway
- 10 programmable parallel demodulation paths
- Pre-configured to support different LoRaWAN regional settings.
- Allow customizing LoRaWAN regional parameters.
- Different kinds of LoRaWAN Connections such as
 - Semtech UDP Packet Forwarder
 - LoRaWAN Basic Station
 - ChirpStack-Gateway-Bridge (MQTT)
- Built-in [ChirpStack](#) local LoRaWAN server
- Built-in [Node-Red](#) local Application server
- Act as Modbus Slave mode

1.5 LED Indicators

MS48-LR has totally four LEDs, They are:

LED 1 (ETH LED): This GREEN LED will blink GREEN when the ETH port is connecting

LED 2 (RET LED): This GREEN LED will show different colors in different states:

SOLID GREEN: When the Long press 4-5s Toggle button, the device will reload the Network and Initialize wifi configuration, This GREEN LED will SOLID GREEN Until the reload is finished.

BLINKING GREEN: When the Long press 10s Toggle button, the device will restore the factory settings, This GREEN LED will BLINKING GREEN Until the restore is finished.

LED 3: This GREEN LED is undefined

LED 4 (Power LED): This GREEN LED will be solid if the device is properly powered

LED 5 (WIFI LED): This LED shows the WIFI interface connection status.

SOLID GREEN: The device enables the WiFi WAN Client and connects to the WiFi successfully.

BLINKING GREEN: The device's WiFi WAN Client connection is unsuccessful.

OFF: The device WiFi WAN Client is not enabled.

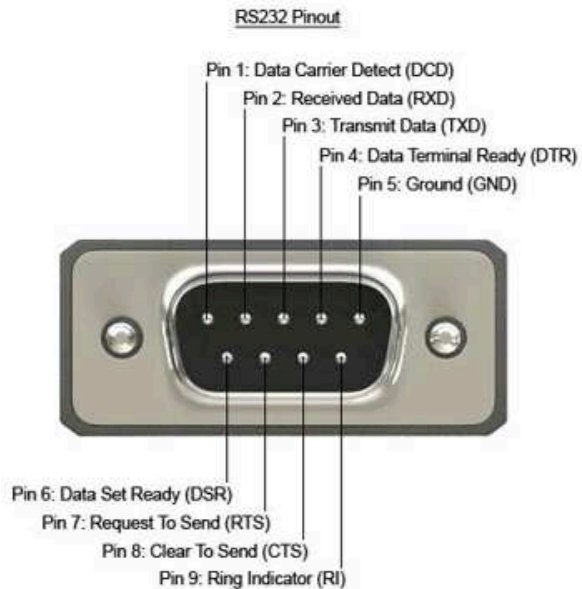
LED 6 (SYS LED): This GREEN LED will show different colors in different states:

SOLID GREEN: The device is alive with a LoRaWAN server connection.

BLINKING GREEN: a) no LoRaWAN Connection. or b) Device is in booting stage, in this stage, it will BLINKING GREEN for several seconds and then with BLINKING GREEN together

OFF: Device doesn't have an Internet connection.

1.6 RS485 & RS232 interface



1.7 Button Instruction

The MS48-LR has two black buttons, which are:

RST: Press and release, and the gateway will restart

LED status: All LEDs are off except for the LED 4 (Power LED).

Toggle:

Long press 4-5s : the gateway will reload the Network and Initialize wifi configuration

LED status: LED 2 (RET LED) will SOLID GREEN Until the reload is finished.

Long press more than 10s: the gateway will restore the factory settings.

LED status: LED 2 (RET LED) will BLINKING GREEN Until the restore is finished.

1.8 Installation



2. Quick Start

The MS48-LR supports network access via Ethernet or Wi-Fi connection and runs without a network.

In most cases, the first thing you need to do is make the MS48-LR accessible to the network.

2.1 Access and Configure MS48-LR

2.1.1 Find IP address of MS48-LR

Method 1: Connect via MS48-LR WiFi

At the first boot of MS48-LR, it will auto generate a WiFi network called *dragino-xxxxxx* with password:

dragino+dragino

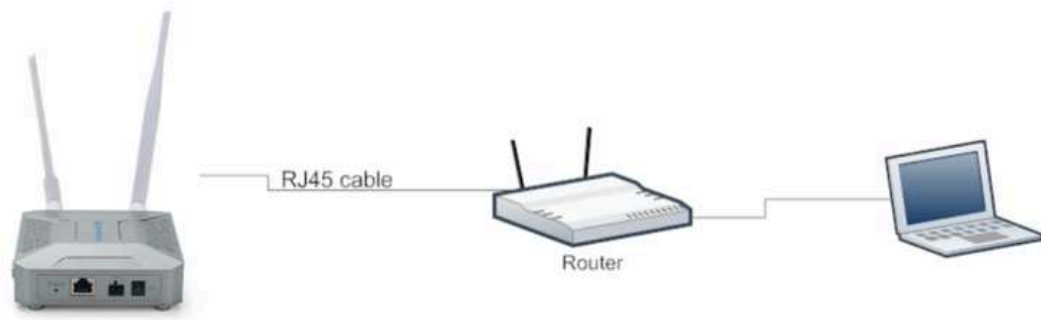
User can use a PC to connect to this WiFi network. The PC will get an IP address 10.130.1.xxx and the MS48-LR has the default IP **10.130.1.1**



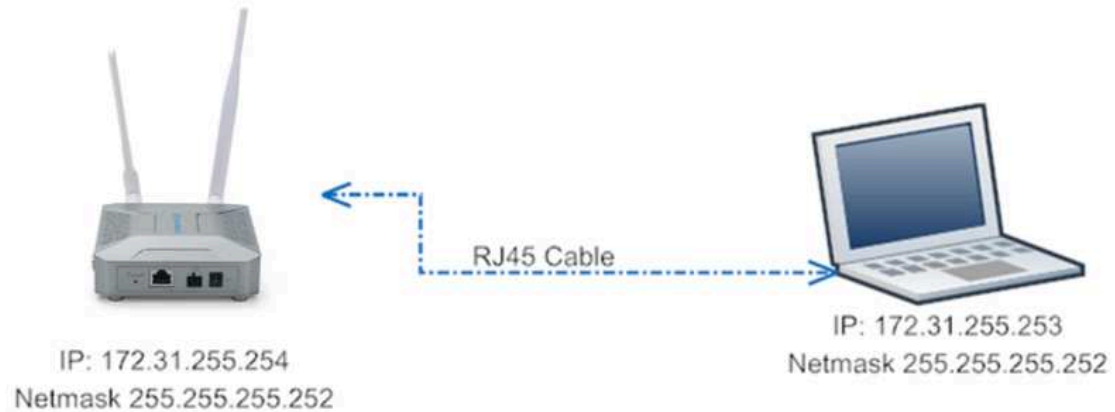
Method 2: Connect via Ethernet with DHCP IP from the router

Connect the MS48-LR Ethernet port to your router and MS48-LR can obtain an IP address from your router. In the router's management portal, you should be able to find what IP address the router has assigned to the MS48-LR.

You can also use this IP to connect.



Method 3: Connect via MS48-LR Fallback IP



Steps to connect via fallback IP:

1. Connect the PC's Ethernet port to MS48-LR's WAN port
2. Configure PC's Ethernet port has IP: 172.31.255.253 and Netmask: 255.255.255.252

Settings --> Network & Internet --> Ethernet --> Change advanced sharing options --> Double-click "Ethernet" --> Internet Protocol Version 4 (TCP/IPv4)

As in the below photo:

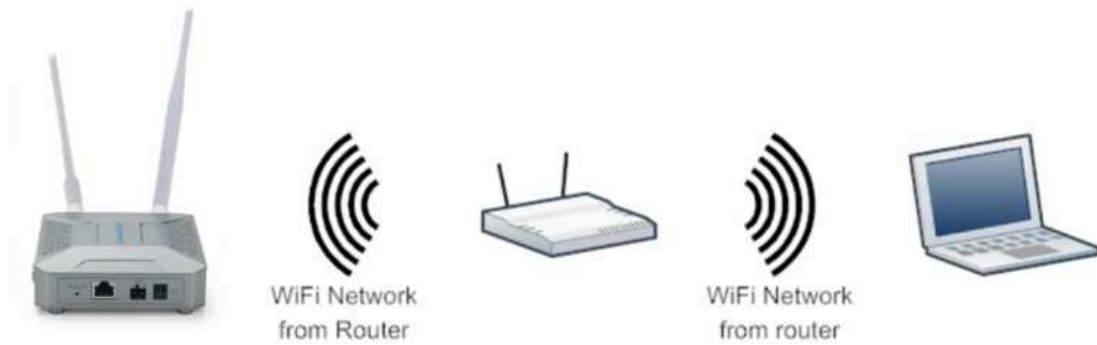
Configure computer Ethernet port steps video: [fallback ip.mp4](#)

If you still can't access the MS48-LR fallback ip, follow this connection to debug : [Trouble Shooting](#)

3. In the PC, use IP address 172.31.255.254 to access the MS48-LR via Web or Console.



Method 4: Connect via WiFi with DHCP IP from the router



Fill in the WiFi information by checking the box and clicking **Save&Apply**

 [LoRa](#) [LoRaWAN](#) [Modbus](#) [Network](#) [System](#) [Server](#) [LogRead](#) [Home](#) [Logout](#)

WiFi Settings

WiFi Mode Wi-Fi WAN Client

WiFi WAN Client Settings

Enable WiFi WAN Client ☒

Host WiFi SSID dragino-RD

Passphrase [Show](#)

WiFi Survey Choose WiFi SSID...

Proto Type DHCP

[Save&Apply](#) [Cancel](#)

Wi-Fi configuration successful



2.1.2 Access Configure Web UI

Web Interface

Open a browser on the PC and type the MS48-LR ip address (depends on your connect method)

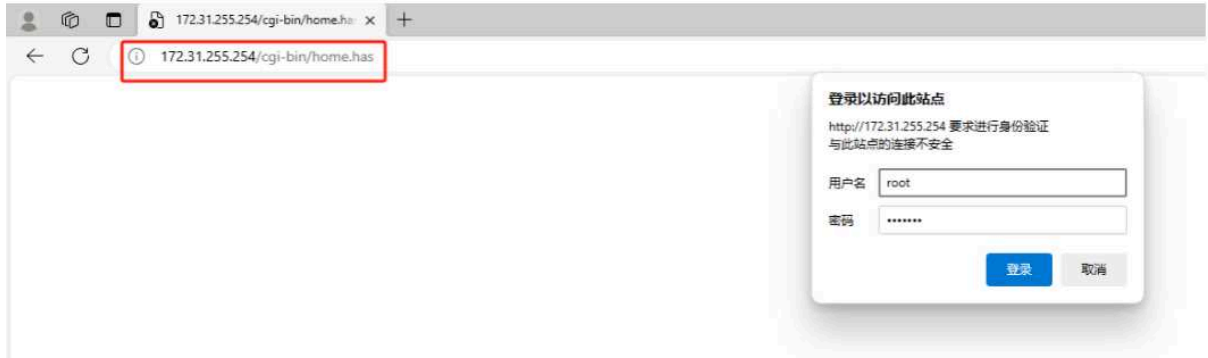
http://IP_ADDRESS or <http://172.31.255.254>(Fallback IP)

You will see the login interface of MS48-LR as shown below.

The account details for Web Login are:

User Name: root

Password: dragino



2.2 Typical Network Setup

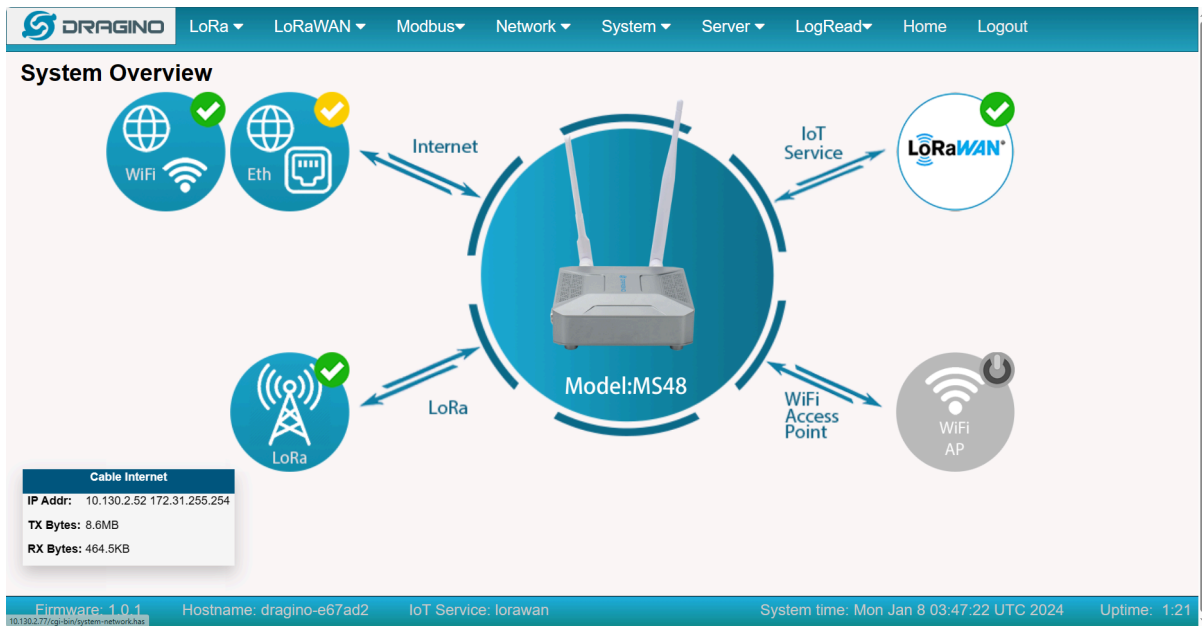
2.2.1 Overview

MS48-LR supports flexible network set up for different environment. This section describes the typical network topology can be set in MS48-LR. The typical network set up includes:

- **WAN Port Internet Mode**
- **WiFi Client Mode**
- **Cellular Mode**

2.2.2 Use the WAN port to access the Internet

By default, the MS48-LR is set to use the WAN port to connect to an upstream network. When you connect the MS48-LR's WAN port to an upstream router, MS48-LR will get an IP address from the router and have Internet access via the upstream router. The network status can be checked on the [home page](#):

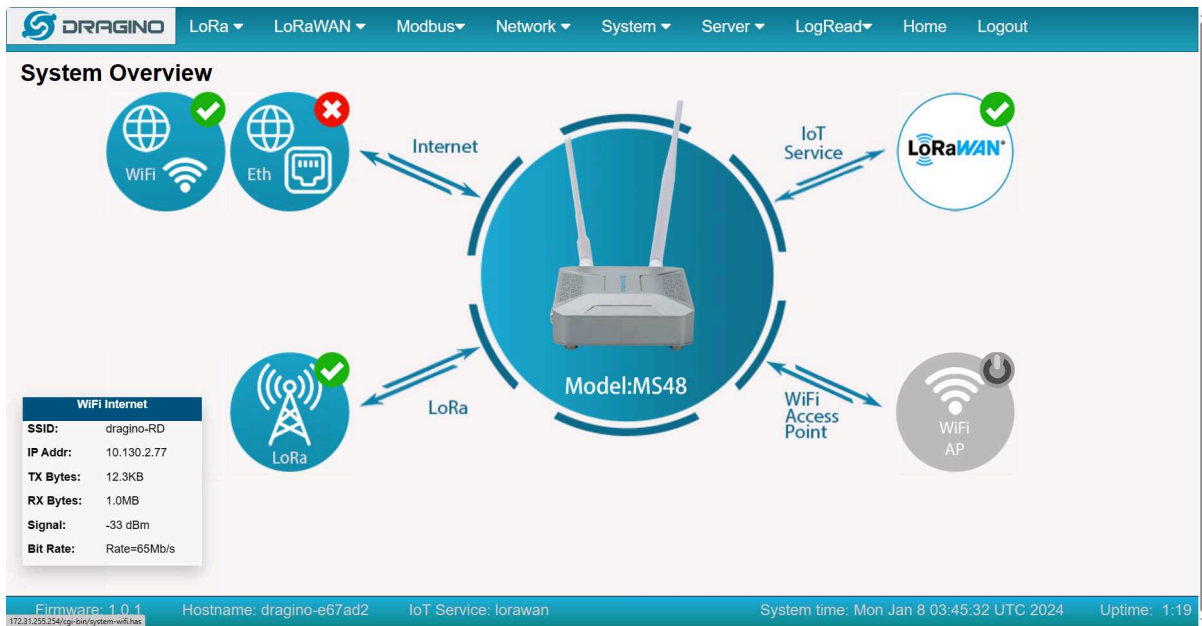


2.2.3 Access the Internet as a WiFi Client

In the WiFi Client Mode, MS48-LR acts as a WiFi client and gets DHCP from an upstream router via WiFi.

The settings for WiFi Client is under page [Network --> Wi-Fi](#)

In the WiFi Survey Choose the WiFi AP, and input the Passphrase then click [Save & Apply](#) to connect.



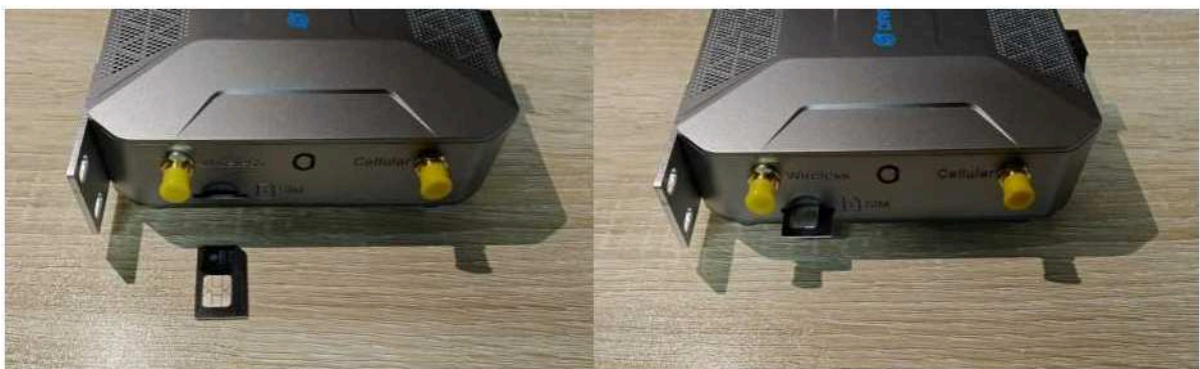
2.2.4 Use built-in 4G modem for internet access

Users can see whether MS48-LR has EC25 on the label of the gateway to determine whether there is 3G/4G Cellular modem.

If the MS48-LR has 3G/4G Cellular modem, user can use it as main internet connection or back up.

First, install the Micro SIM card as below direction

Second, Power off/ ON MS48-LR to let it detect the SIM card.



The set up page is [Network --> Cellular](#)

While use the cellular as Backup WAN, device will use Cellular for internet connection while WAN port or WiFi is not valid and switch back to WAN port or WiFi after they recover.

2.2.5 Check Internet connection

In the [Home](#) page, we can check the Internet connection.

- GREEN Tick : This interface has Internet connection.
- Yellow Tick : This interface has IP address but don't use it for internet connection.
- RED Cross : This interface doesn't connected or no internet.



2.3 Bridge LoRaWAN network to Modbus network

By following the steps below in the configuration example, Users can convert the uplink data of the lorawan sensor to modbus RTU/TCP data

MS48-LR eliminates the need to add additional lorawan gateways to implement [LoRaWAN to Modbus RTU/TCP](#)

Step 1: Configure the LoRa Radio to your area Frequency Plan

The Frequency Plan has to be set the same as the Sensor node Frequency Plan.

The screenshot shows the 'LoRa Configuration' page of a DRAGINO web interface. The top navigation bar includes links for LoRa, LoRaWAN, Modbus, Network, System, Server, LogRead, Home, and Logout. The main configuration area includes a 'Debug Level' dropdown set to 'Low'. Under 'Radio Settings', the 'Stat Package Period (sec)' is set to 30. The 'Frequency Plan' dropdown menu is open, displaying a list of options: EU868 Europe 868Mhz (863~870), CN470 China 470MHz (470~510), US915 United States 915Mhz (902~928), AU915 Australia 915Mhz (915~928), IN865 India 865Mhz (865~867), KR920 Korea 920MHz (920~923), AS923 Asia AS923-1 (920-923MHz), AS923 Asia AS923-1 (923-925MHz), AS923 Asia AS923-2 (922-923MHz), AS923 Asia AS923-3 (916-918MHz), AS923 Asia AS923-4 (917-919MHz), RU864 Russia 864MHz (864~870), and Customized Bands. The 'Static GPS coordinates' section has a red question mark icon. It includes an 'Enable Static GPS' checkbox, a 'Latitude' input field with the value 114.240000, and a 'Longitude' input field with the value 450. The 'Current Mode' is set to 'LoRaWAN Semtec'. At the bottom, there are 'Save&Apply', 'Disable', and 'Cancel' buttons.

Step 2: Copy the unique Gateway EUI & Configure the LoRaWAN Server address

Users need to connect the MS48-LR configuration to the built-in lorawan server.

Every MS48-LR has a unique gateway EUI. The ID can be found on the LoRaWAN Semtech page:

DRAGINO

LoRa ▼ LoRaWAN ▼ Modbus ▼ Network ▼ System ▼ Server ▼ LogRead ▼ Home Logout

LoRaWAN Configuration

LoRaWAN Semtech UDP

LoRaWAN Basic Station

General Settings

Email

dragino-106e6e@dragino.com

Gateway EUI

02817bdfc106e6e

Primary LoRaWAN Server

Service Provider

Local Host/Built-in Server ▼

Uplink Port

1700

Downlink Port

1700

Secondary LoRaWAN Server

Service Provider

Disable ▼

Packet Filter

Primary server Fport Filter ?

Level 0 ▼

DevAddr Filter ?

Level 0 ▼

Secondary server Fport Filter

Level 0 ▼

DevAddr Filter

Level 0 ▼

Add Filter

Server Name:

▼

Filter type:

▼

Filter Value

1,2,3 or 018229B

ADD_FILTER

DELET Filter

▼

DELETE

Current Mode:LoRaWAN Semtech UDP

Save&ApplyCancel

Step 1: Copy Gateway EUI

Setp 2: Select to the Local host

Select 3: Click the "Save&Apply" button

Step 3: Enable the Built-in LoRaWAN Network Server

DRAGINO

LoRaLoRaWANModbusNetworkSystemServerLogReadHomeLogout

Built-In Network Server

Network Server

Application Server

Platform Provider

Chirpstack

Enable☒Launch

Service Status

Running

Server Versions

Chirpstack V4

Frequency Plan

Frequency Sub Band

Management

Save&Apply

US915

EU433

CN470

CN779

RU864

IN865

EU868

US915

AU915

KR920

AS923

Step 1: Check the Enable checkbox

Step 2: Select the Frequencct Plan

Stpe 3: Click the "Save&Apply" button

Step 4: Logging to the Built-in LoRaWAN Network Server

Login address: <http://<Gateway Ip>:8080>
For example: <http://10.130.1.1:8080>

ChirpStack login

* Username / email:

* Password:

👁

Submit

Step 5: Register the gateway to the built-in ChirpStack

Copy Gateway EUI from the previous step to the following interface:

ChirpStack

ChirpStack

Network Server

Dashboard

Tenants

Users

API Keys

Device Profile Templates

Regions

Tenant

Dashboard

Users

API Keys

Device Profiles

Gateways

Applications

Tenants / ChirpStack / Gateways / Add

Add gateway

General

Tags

Metadata

* Name

Dragino-Test

Description

* Gateway ID (EUI64)

02817bfdfef106e0e

MSB

* Stats interval (secs)

30

Location

+

-

3) .Submit Add

Step 6: Register the Sensor-node to the built-in ChirpStack

The MS48-LR gateway is already set up to connect to the built-in ChirpStack network, so we now need to configure the sensor device to connect to the built-in ChirpStack.

First we need to add the Device Profiles and Application (ignore them if they are already done).

1) Add Device Profiles

The screenshot shows the ChirpStack web interface for adding a device profile. The left sidebar contains a navigation menu with options like Dashboard, Tenants, Users, API Keys, Device Profile Templates, Regions, and Tenant. The main content area is titled 'Add device profile' and has tabs for General, Join (OTAA / ABP), Class-B, Class-C, Codec, Relay, Tags, and Measurements. The 'General' tab is active. The form contains several fields: 'Name' (with a red box and annotation '1. Enter a name'), 'Description', 'Region' (with a red box and annotation '2. Select the same region'), 'MAC version' (with a red box and annotation '3. Select the 1.0.3'), 'Regional parameters revision', 'ADR algorithm', 'Flush queue on activate' (a toggle switch), 'Expected uplink interval (secs)' (a text input with '3600'), and 'Device-status re' (a text input with '1'). A blue 'Submit' button is at the bottom left, with a red box and annotation '4. Click the Submit button'.

ChirpStack

Search...

ChirpStack

Network Server

Dashboard

Tenants

Users

API Keys

Device Profile Templates

Regions

Tenant

Dashboard

Users

API Keys

Device Profiles

Gateways

Applications

Add device profile

General Join (OTAA / ABP) Class-B Class-C Codec Relay Tags Measurements

* Name
Dragino-Test **1. Enter a name**

Description

* Region
EU868 **2. Select the same region**

Region configuration ⓘ
eu868

* MAC version ⓘ
LoRaWAN 1.0.3 **3. Select the 1.0.3**

* Regional parameters revision ⓘ
A

* ADR algorithm ⓘ
Default ADR algorithm (LoRa only)

Flush queue on activate ⓘ
☒

* Expected uplink interval (secs) ⓘ
3600

Device-status re
1

Submit **4. Click the Submit button**

2) Add End Node Device

Create an Application

ChirpStack

Network Server

Dashboard

Tenants

Users

API Keys

Device Profile Templates

Regions

Tenant

Dashboard

Users

API Keys

Device Profiles

Gateways

Applications

Tenants / ChirpStack / Applications / Add

Add application

* Name

Dragino-Test

Description

Submit

Add a device for the sensor node

ChirpStack

Search...

admin

ChirpStack

Network Server

Dashboard

Tenants

Users

API Keys

Device Profile Templates

Regions

Tenant

Dashboard

Users

API Keys

Device Profiles

Gateways

Applications

EU868 application id: 40079201-0c7c-473f-9300-1243210e10c0

Delete application

Devices

Multicast groups

Relays

Application configuration

Integrations

Add device

Selected devices

☐	Last seen	DevEUI	Name	Device profile	Battery
☐	2023-11-09 09:24:23	70b3d57ed0051e66	147852	EU868	
☐	2023-11-09 09:24:28	70b3d57ed0051e22	70 B3 D5 7E D0 05 1E 22	EU868	
☐	2023-11-08 15:22:45	70b3d57ed0051e33	70 B3 D5 7E D0 05 1E 33	EU868	
☐	2023-11-09 09:24:28	70b3d57ed0051e77	70 B3 D5 7E D0 05 1E 77	EU868	
☐	2023-09-14 16:26:52	aaaaaaaaaaaa	LHT	EU868	
☐	2023-08-24 19:44:20	6081f9102d23eda8	Isn50v2	EU868	

<

1

>

10 / page

Enter Device EUI, Join EUI(APP EUI)and APPKEY of the node Device, and select the Device profile added in the previous step

ChirpStack

Network Server

Dashboard

Tenants

Users

API Keys

Device Profile Templates

Regions

Tenant

Dashboard

Users

API Keys

Device Profiles

Gateways

Applications

Tenants / ChirpStack / Applications / Dragino-Test / Add device

Add device

Device

Tags

Variables

Name

Dragino-Test

1. Enter a name

Description

Device EUI (EUI64)

2. Enter Sensor node's DEUI

Join EUI (EUI64)

3. Enter Snesor Node JEUI

Device profile

Dragino-Test

3. Select an device profile

Device is disabled

Disable frame-counter validation

Submit

5. Submit

ChirpStack

Search...

ChirpStack

Network Server

Dashboard

Tenants

Users

API Keys

Device Profile Templates

Regions

Tenant

Dashboard

Users

API Keys

Device Profiles

Gateways

Applications

Tenants / ChirpStack / Applications / Dragino-Test / Devices / Dragino-Test

Dragino-Test device eui: bcaed81560cdfbb7

Dashboard Configuration OTAA keys Activation Queue Events LoRaWAN frames

* Application key ⓘ

Submit

6. Enter the Sensor node's APP key

7. Submit

Step 7: Configure Modbus RTU/TCP Slave

The gateway can act as a Modbus RTU slave to run, the user can set a range of the register to write a sensor node uplink data.

For example, there is a sensor node EUI is 70b3d57ed0051e22, the register start is 0xABCD and the register length is 14, which means the uplink data will be written to the register starting at 0xABCD register and the maximum write length not to exceed 14 registers.

So the sensor node 70b3d57ed0051e22 uplink data will be written to the 0xABCD - 0xABE1 register.

Note: Since the length of the payload is the same for different sensor nodes if The length of the data is greater than the configuration length, the data will be replaced with FFFF.

Slave Setting:

Slave Mode ---> Support both RTU and TCP mode

Enable Modbus Slave ---> Enable Slave

Slave Address/Port ---> Set the slave address/port

Show Sensor History ---> See the sensor data log

Sensor Settings:

Device EUI ---> Sensor node's EUI

Register Start ---> Setting the start register address to write the sensor node's data

Register Length ---> The maximum write length with register

Note: Only 1 sensor can be written in the range set by the registers

1) RTU Mode:

 LoRa ▼ LoRaWAN ▼ Modbus ▼ Network ▼ System ▼ Server ▼ LogRead ▼ Home Logout

Modbus RTU/TCP -- RS485

RS485 RTU/TCP

Slave Settings

Salve Mode

Modbus RTU ▼

Enable Modbus Slave

☒

Slave Address

17

[Show Sensor History](#)

Save&Apply

Reload

Sensor Settings

Device EUI

Device EUI

Register Start

Hexadecimal: 0000

Register Length

12

ADD

Delete Device

70b3d57ed0051e22 ▼

DELETE

ID	Salve Address	Register Address(Start)	Register Legth	Device EUI
1		ABCD	14	70b3d57ed0051e22
2		1231	15	70b3d57ed0051e66
---	---	---	---	---

After the sensor node is active at the built-in server Chirpstack, the user can add it to this page and enable the Modbus RS485-RTU Slave.

Then MS48-LR will write the uplink data to the 03 code register and record the uplink data.

PLC read the MS48-LR register

Settings:

Function : 03 code

Bit rate : 9600

Parity bit : none

Stop bit : 1

Response Timeout : Greater than 3000ms

2) TCP Mode:

DRAGINO

LoRa ▾ LoRaWAN ▾ Modbus ▾ Network ▾ System ▾ Server ▾ LogRead ▾ Home Logout

Modbus RTU/TCP -- RS485

RS485 RTU/TCP

Slave Settings

Slave Mode

Modbus TCP/IP ▾

Enable Modbus Slave

☒

Slave Port

502

[Show Sensor History](#)

Save&Apply

Reload

Sensor Settings

Device EUI

Device EUI

Register Start

Hexadecimal: 0000

Register Length

12

ADD

Delete Device

70b3d57ed0051e22 ▾

DELETE

ID	Slave Address	Register Address(Start)	Register Legth	Device EUI
1		ABCD	14	70b3d57ed0051e22
2		1231	15	70b3d57ed0051e66
---	---	---	---	---

PLC read the MS48-LR register

Settings:

Function : 03 code

IP Address : Gateway IP Address

Server Port : Slave Port

Connect Timeout : Greater than 3000ms

3) PLC(Modbus server/master) data show:

Modbus Poll - Mbpoll1

File Edit Connection Setup Functions Display View Window Help

05 06 15 16 17 22 23 TC ?

Mbpoll1

Tx = 55: Err = 1: ID = 17: F = 03: SR = 1500ms

	Alias	43980	Alias	44000
0		0x0000		0x0000
1		0x017B		0x0000
2		0x8D19		0x0000
3		0x3C1A		0x0000
4		0x007B		0x0000
5		0xCCE8		0x0000
6		0x7FFF		0x0000
7		0x7FFF		0x0000
8		0x017F		0x0000
9		0xFF7F		0x0000
10		0xFF00		
11		0x654E		
12		0xEAA4		
13		0x0000		
14		0x0000		
15		0x0000		
16		0x0000		
17		0x0000		
18		0x0000		
19		0x0000		

Dev Address

Fcnt

RSSI

Payload

Date

History Data:

Click the Show Sensor History will launch to this page


LoRa ▾
LoRaWAN ▾
Modbus ▾
Network ▾
System ▾
Server ▾
LogRead ▾
Home
Logout

Listen List

Selection

70b3d57ed0051e22 ▾

Search

Listen Frames

Count	Slave Address	Dev EUI	Dev Address	Fcnt	RSSI	Payload	Date
15543	17	70b3d57ed0051e22	017b8d19	14340	007c	cce87fff7fff017fff7fff	654ed220
15544	17	70b3d57ed0051e22	017b8d19	14354	007c	cce87fff7fff017fff7fff	654ed274
15545	17	70b3d57ed0051e22	017b8d19	14355	007c	cce87fff7fff017fff7fff	654ed27a
15546	17	70b3d57ed0051e22	017b8d19	14356	007c	cce87fff7fff017fff7fff	654ed280
15547	17	70b3d57ed0051e22	017b8d19	14357	007d	cce87fff7fff017fff7fff	654ed286
15548	17	70b3d57ed0051e22	017b8d19	14379	007c	cce87fff7fff017fff7fff	654ed30a
15549	17	70b3d57ed0051e22	017b8d19	14381	007a	cce87fff7fff017fff7fff	654ed316
15550	17	70b3d57ed0051e22	017b8d19	14398	007c	cce87fff7fff017fff7fff	654ed37c
15551	17	70b3d57ed0051e22	017b8d19	14405	007a	cce87fff7fff017fff7fff	654ed3a6
15552	17	70b3d57ed0051e22	017b8d19	14406	007d	cce87fff7fff017fff7fff	654ed3ac
15553	17	70b3d57ed0051e22	017b8d19	14411	007c	cce87fff7fff017fff7fff	654ed3ca
15554	17	70b3d57ed0051e22	017b8d19	14419	007c	cce87fff7fff017fff7fff	654ed3fa
15555	17	70b3d57ed0051e22	017b8d19	14421	007a	cce87fff7fff017fff7fff	654ed406
15556	17	70b3d57ed0051e22	017b8d19	14427	007c	cce87fff7fff017fff7fff	654ed42a
15557	17	70b3d57ed0051e22	017b8d19	14428	007c	cce87fff7fff017fff7fff	654ed430
15558	17	70b3d57ed0051e22	017b8d19	14435	007c	cce87fff7fff017fff7fff	654ed45a
15559	17	70b3d57ed0051e22	017b8d19	14440	007b	cce87fff7fff017fff7fff	654ed478
15560	17	70b3d57ed0051e22	017b8d19	14450	007d	cce87fff7fff017fff7fff	654ed4b4

Step 8: Configure Sensor decode to Built-in Chirpstack

1) Add Sensor's decode to Chirpstack

Users can find the ChirpStack v4 decoder code for the Dragino End node in this [link](#):

<https://github.com/dragino/dragino-end-node-decoder>

The following example is to add the LHT65N decoder:

ChirpStack

Network Server

Dashboard
Tenants
Users
API Keys
Device Profile Templates
Regions

Tenant

Dashboard
Users
API Keys
Device Profiles
Gateways
Applications

Tenants / ChirpStack / Device profiles / LHT65N

LHT65N device profile id: 25f7c47e-eed2-494d-9bc8-3833928876c8

General Join (OTAA / ABP) Class-B Class-C **Codec** Relay Tags Measurements

Payload codec

JavaScript functions

None

Cayenne LPP

JavaScript functions

```
3 data: Decode(input.fPort, input.bytes, input.variables)
4 };
5 }
6
7 function Str1(str2) {
8   var str3 = "";
9   for (var i=0; i<str2.length; i++) {
10     if (str2[i]<=0x0f) {
11       str2[i] = "0" + str2[i].toString(16) + "";
12     }
13     str3 += str2[i].toString(16) + "";
14   }
15   return str3;
16 }
17 function str_pad(byte) {
18   var zero = "00";
19   var hex = byte.toString(16);
20   var tmp = 2 - hex.length;
21   return zero.substr(0, tmp) + hex + "";
22 }
23 function datalog(i, bytes) {
```

1. Click the "Codec"

2. Select the "JavaScript functions"

ChirpStack

Network Server

- Dashboard
- Tenants
- Users
- API Keys
- Device Profile Templates
- Regions

Tenant

- Dashboard
- Users
- API Keys
- Device Profiles
- Gateways
- Applications

Payload codec

JavaScript functions 3. Copy the LHT65N Chirpstack decoder code to the Chirpstack Codec

Codec functions

```
18 var hex= byte.toString(16);
19 var tmp = 2-hex.length;
20 return zero.substr(0,tmp) + hex + " ";
21 }
22
23 function datalog(i,bytes) {
24 var Ext= bytes[0]&0x0F;
25 var bb;
26 if((Ext=='1')||(Ext=='9'))
27 {
28 bb=parseFloat(((bytes[0]<<24)>>16 | bytes[1+i])/100).toFixed(2);
29 }
30 else if(Ext=='2')
31 {
32 bb=parseFloat(((bytes[0]<<24)>>16 | bytes[1+i])/100).toFixed(2);
33 }
34 else if(Ext=='4')
35 {
36 var Exti_pin_level=bytes[0+i] ? "High":"Low";
37 var Exti_status=bytes[1+i] ? "True":"False";
38 bb=Exti_pin_level+Exti_status;
39 }
40 else if(Ext=='5')
41 {
42 bb=bytes[0+i]<<8 | bytes[1+i];
43 }
44 else if(Ext=='6')
```

Submit

4. After the configure ,click the "Submit"

2) Check the decode on ChirpStack

The screenshot displays the ChirpStack web interface for a specific device, LHT65N (EUI: a840411be186e411). The 'Events' tab is selected, showing a list of uplink messages. A red box labeled '1.' highlights the 'Events' tab, and another red box labeled '2.' highlights the 'up' button for the first event.

The decoded data for the selected event is shown on the right side of the interface:

```
tenantName: "ChirpStack"
applicationId: "a35eabc0-d345-422a-94d5-ed9b94411ac1"
applicationName: "LHT65N"
deviceProfileId: "25f7c47e-eed2-494d-9bc8-3833928876c8"
deviceProfileName: "LHT65N"
deviceName: "LHT65N"
devEui: "a840411be186e411"
deviceClassEnabled: "CLASS_A"
tags: {}
devAddr: "01c3eabb"
adr: true
dr: 5
fCnt: 68
fPort: 2
confirmed: false
data: "y9qJfw//AX//f/8="
object: {}
  Ext_sensor: "Temperature Sensor"
  Bat_status: 3
  BatV: 3.032
  Hum_SHT: 63.9
  TempC_SHT: 24.31
  TempC_DS: 327.67
rxInfo: {}
  0: {}
    gatewayId: "02818ffdf6e67ad2"
    uplinkId: 37447
    time: "2024-03-19T06:46:45.818475+00:00"
    rssi: -128
    snr: -5.5
    channel: 4
    location: {}
    context: "64G22g=="
  metadata: {}
    region_common_name: "EU868"
    region_config_id: "eu868"
    crcStatus: "CRC_OK"
txInfo: {}
  frequency: 867300000
  modulation: {}
  lora: {}
```

Step 9: Configure Modbus RTU/TCP Slave Sensor Decode

For example, there is a sensor node EUI is a840411be186e411, the register start is 0x0000 and the register length is 14, which means the uplink data will be written to the register starting at 0x0000 register and the maximum write length not to exceed 14 registers.

Add Decode Data TempC_SHT, the register start address is 0x000F and the register length is 2, which means the Decode Data TempC_SHT will be written to the register starting at 0x000F register and the maximum write length not to exceed 2 registers.

So the sensor node a840411be186e411 uplink data will be written to the 0x0000 - 0x000E register, and the Decode Data TempC_SHT will be written to the 0x000F - 0x0010 register.

Note: Since the length of the payload is the same for different sensor nodes if The length of the data is greater than the configuration length, the data will be replaced with FFFF.

Decode Data:

Device Type	--->	Sensor decoder data
Register Start	--->	Setting the start register address to write the sensor node's decoder data
Register Length	--->	The maximum write length with register

Note: Only 1 Data can be written in the range set by the registers


LoRa LoRaWAN Modbus Network System Server LogRead Home Logout

Device EUI Register Start Register Length

Delete Device

ID	Salve Address	Register Address(Start)	Register Legth	Device EUI
1	13	0	14	a840411be186e411
2	13	1A	14	ffffffff22ffff11
3	13	30	14	ffb3d57ed0051e66
---	---	---	---	---

Decode Data

Device Search

Decode Data Register Start Register Length

Delete Device

ID	Data Type	Register Address(Start)	Register Legth	Device EUI
1	TempC_SHT	F	2	a840411be186e411
2	Hum_SHT	11	2	a840411be186e411
3	BatV	13	2	a840411be186e411
4	TempC_DS	15	2	a840411be186e411

PLC(Modbus server/master) data show:

Note: Since the decoded data is of floating point type, it is converted to an integer before being written to the Mobdbus' registers.

For example: 23.20(real data) ---> 2320[Dec]=0x08FE[Hex](register show).

When the decoded data is negative, the data will be added to 0x10000

For example: -23.20(real data) ---> -2320+0x10000=63216[Dec]=0xF6F0[Hex](register show).

Modbus Poll - Mbpoll1.mbp

File Edit Connection Setup Functions Display View Window Help

05 06 15 16 17 22 23 TC ? ?

Mbpoll1.mbp

Tx = 277: Err = 0: ID = 13: F = 03: SR = 2000ms

	Alias	00000	Alias	00020	Alias
0		0x001C		0x0000	
1	Dev Address	0x2B7D		0x7FFF	
2	Fcnt	0x0042		0x0000	
3	RSSI	0x0082		0x0000	
4		0xCBD5	TempC_DS	0x0000	
5		0x08B3		0x0000	
6	Payload	0x035C		0x0113	
7		0x017F		0xACE5	
8		0xFF7F		0x027B	
9		0xFF00		0x006D	
10	Date	0x65F4		0xCCDD	
11		0xF142		0x7FFF	
12		0x0000		0x7FFF	
13		0x0000	TempC_SHT	0x017F	
14		0x0000		0xFF7F	
15		0x08B3	Hum_SHT	0xFF00	
16		0x0000		0x65F4	
17		0x0056	BatV	0xF1B3	
18		0x0000		0x0000	
19		0x0BD5		0x0000	

2.4 Accept data to registers for specified Fport

For example, there is a sensor node EUI is f4bbf5a0da6f4da5,

- the sensor payload uplink is using Fport=2
- the sensor status uplink is using Fport=5

If the accept fport is not set, it may cause the registers to be written with 0 decoded data during status uplink, because status uplinks typically do not contain decoded data.

To avoid 0 data being written to the registers. the users can set accept Fport for the sensor,
i.e., the data will be written to the register only when the MS48-LR receives the uplink of the specified Fort

Accept Fport:

DevEUI ---> Sensor node DevEUI

Accept FPort ---> Setting the Accept Fport

Accept FPort

Device Search

Dev EUI

Accept FPort

Option

ADD Accept FPort

Delete Accept FPort

Register Address

DELETE Accept FPort

ID	Device EUI	Accept FPort	---	---	---
1	f4bbf5a0da6f4da5	2	---	---	---
---	---	---	---	---	---

PLC(Modbus server/master) data show:

Modbus Poll - Mbpoll2
File Edit Connection Setup Functions Display View Window Help
05 06 15 16 17 22 23 TC ? ?
Mbpoll2
Tx = 706: Err = 0: ID = 1: F = 03: SR = 1000ms

	Alias	00000	Alias	00010	Alias	00020	Alias
0		0x0140		0x6661		0x0000	
1		0x724F	Date	0x6885	air_pressure	0x273C	
2	Fcnt ←	0x01AB		0x0000	humidity	0x0243	
3	RSSI ←	0x006E		0x0000	BatV	0x0DF8	
4		0x0DF8		0x0000	Sensor Decoder Data	0x0000	
5		0x0102		0x0000		0x0000	
6	Fport=2	0x0243		0x0000		0x0000	
7	Payload	0x273C		0x0000		0x0000	
8		0x033C		0x0000		0x0000	
9		0x0000		0x0000		0x0000	

Modbus Poll - Mbpoll2

File Edit Connection Setup Functions Display View Window Help

05 06 15 16 17 22 23 TC ? ?

Mbpoll2

Tx = 928: Err = 0: ID = 1: F = 03: SR = 1000ms

	Alias	00000	Alias	00010	Alias	00020	Alias
0	Dev Address	0x0140		0x0000		0x0000	
1		0x724F		0x0000	air_pressure	0x273B	
2	Fcnt	0x01AD		0x0000	humidity	0x023B	
3	RSSI	0x0069		0x0000	BatV	0x0DFE	
4		0x3701		0x0000		0x0000	
5	Fort=5	0x0001		0x0000	sensor Decode	0x0000	
6	Payload	0xFF0D		0x0000	Data	0x0000	
7		0xFE00		0x0000		0x0000	
8		0x6661		0x0000		0x0000	
9	Date	0x6AC6		0x0000		0x0000	

3. Web Configure Pages

3.1 Home

Shows the system running status:

3.2 LoRa Settings

3.2.1 LoRa --> LoRa

This page shows the LoRa Radio Settings. There is a set of default frequency bands according to LoRaWAN protocol, and users can customize the band* as well.

Different MS48-LR hardware versions can support different frequency ranges:

- **868**: valid frequency: 863Mhz ~ 870Mhz. for bands EU868, RU864, IN865, or KZ865.
- **915**: valid frequency: 902Mhz ~ 928Mhz. for bands US915, AU915, AS923 or KR920

After the user choose the frequency plan, the user can see the actual frequency is used by checking the page **LogRead --> LoRa Log**


LoRa ▼ LoRaWAN ▼ Modbus ▼ Network ▼ System ▼ Server ▼ LogRead ▼ Home Logout

LoRa Configuration

Debug Level

Radio Settings

Stat Package Period (sec)

Frequency Plan

Static GPS coordinates ?

Enable Static GPS

Latitude

Current Mode: **LoRaWAN Semtech**

EU868 Europe 868Mhz (863~870)
CN470 China 470MHz (470~510)
US915 United States 915Mhz (902~928)
AU915 Australia 915Mhz (915~928)
IN865 India 865Mhz (865~867)
KR920 Korea 920MHz (920~923)
AS923 Asia AS923-1 (920~923MHz)
AS923 Asia AS923-1 (923~925MHz)
AS923 Asia AS923-2 (922~923MHz)
AS923 Asia AS923-3 (916~918MHz)
AS923 Asia AS923-4 (917~919MHz)
RU864 Russia 864MHz (864~870)
Customized Bands

Note *: See this instruction for how to customize the frequency band: [How to customized LoRaWAN frequency band - DRAGINO](#)

3.3 LoRaWAN Settings

3.3.1 LoRaWAN --> LoRaWAN Semtech UDP

This page is for the connection set up to a general LoRaWAN Network server such as [TTN](#), [ChirpStack](#), etc.

 DRAGINO

LoRa

LoRaWAN

Modbus

Network

System

Server

LogRead

Home

Logout

LoRaWAN Configuration

General Settings

Email

dragino-106e6e@dragino.com

Gateway EUI

02817bdfef106e6e

Primary LoRaWAN Server

Service Provider

The Things Network V3

Server Address

eu1.cloud.thethings.network

Uplink Port

1700

Downlink Port

1700

Secondary LoRaWAN Server

Service Provider

Local Host/Built-in Server

Uplink Port

1700

Downlink Port

1700

Packet Filter

Primary server Fport Filter ?

Level 0

DevAddr Filter ?

Level 0

Secondary server Fport Filter

Level 0

DevAddr Filter

Level 0

Add Filter

Server Name:

Filter type:

Filter Value

1,2,3 or 018229B

ADD_FILTER

DELET Filter

DELETE

Current Mode:LoRaWAN Semtech UDP

Save&Apply

Cancel

3.3.2 LoRaWAN --> LoRaWAN Basic Station

This page is for the connection set up to the TTN Basic Station, AWS-IoT, etc.

 DRAGINO

LoRa ▾LoRaWAN ▾Modbus ▾Network ▾System ▾Server ▾LogRead ▾HomeLogout

LoRaWAN -- Basic Station

LoRaWAN Semtech UDPLoRaWAN Basic Station

General Settings

Email

dragino-106e6e@dragino.com

Gateway ID

02817bdfef106e6e

Restore ?

Restore_Configuration

Primary LoRaWAN Server

Service Provider

Amazon IoT -- Basic Station ▾

CUPS URI

example: https://xxxxxxx.cups.lorawan.us-east-1.amazonaws.com:443

LNS URI

example: wss://xxxxxxx.lns.lorawan.us-east-1.amazonaws.com:443

CUPS trust

Not Found

选择文件未选择文件

Upload_CUPS_Trust

LNS trust

Not Found

选择文件未选择文件

Upload_LNS_Trust

Private key

Not Found

选择文件未选择文件

Upload_Private_key

Cert pem

Not Found

选择文件未选择文件

Upload_Cert_pem

Current Mode: **LoRaWAN Semtech UDP** Click Save & Apply will change to mode: **LoRaWAN Basic Station**

Save&ApplyCancel

Please see this instruction to know more detail and a demo for how to use of LoRaWAN Basic Station: [Use of LoRaWAN Basic Station - DRAGINO](#)

3.4 Network Settings

3.4.1 Network --> WiFi

Users can configure the wifi WAN and enable Wifi Access Point on this interface

 DRAGINO

LoRa ▾

LoRaWAN ▾

Modbus ▾

Network ▾

System ▾

Server ▾

LogRead ▾

Home

Logout

WiFi Settings

WiFi Mode

Wi-Fi WAN Client ▾

Wi-Fi Access Point

Wi-Fi WAN Client

WiFi WAN Client Settings

Enable WiFi WAN Client ☒

Host WiFi SSID

dragino-RD

WiFi Survey

Choose WiFi SSID... ▾

Passphrase

.....

Show

Proto Type

DHCP ▾

Save&Apply

Cancel

Network

Wi-Fi

Cellular

Network Status

3.4.2 Network --> System Status

 LoRa ▼ LoRaWAN ▼ Modbus ▼ Network ▼ System ▼ Server ▼ LogRead ▼ Home Logout

Network Status

Network / WiFi Status

```
wlan0: connected to dragino-RD
"Ralink MT7601U"
wifi (mt7601u), 34:7D:E4:BC:38:27, hw, mtu 1500
ip4 default
inet4 10.130.2.24/24
route4 10.130.2.0/24 metric 600
route4 default via 10.130.2.1 metric 600
inet6 fe80::b1ed:a4f7:8275:7de7/64
route6 fe80::/64 metric 1024

br-43540c0229a7: connected (externally) to br-43540c0229a7
"br-43540c0229a7"
bridge, 02:42:CB:B7:00:45, sw, mtu 1500
inet4 172.18.0.1/16
route4 172.18.0.0/16 metric 0
inet6 fe80::42:cbff:feb7:45/64
route6 fe80::/64 metric 256

docker0: connected (externally) to docker0
"docker0"
bridge, 02:42:6C:C9:EA:07, sw, mtu 1500
inet4 172.17.0.1/16
route4 172.17.0.0/16 metric 0

eth0: connecting (getting IP configuration) to Wired connection 1
"eth0"
ethernet (dwmac-sun8i), 02:81:7B:10:6E:6E, hw, mtu 1500

fallback: disconnected
"fallback"
macvlan, DE:BD:F0:3E:27:6C, sw, mtu 1500

veth61b0bb9: unmanaged
"veth61b0bb9"
ethernet (veth), CE:C9:B3:0E:F1:91, sw, mtu 1500
```

Refresh

3.4.3 Network --> Network

In the **Network --> Network** interface, Users can set the Ethernet WAN static ip address.


LoRa ▼ LoRaWAN ▼ Modbus ▼ Network ▼ System ▼ Server ▼ LogRead ▼ Home Logout

Network

Ethernet WAN Settings

Mode Static ▼

IP Address

Gateway

Netmask

DNS

Save&Apply

3.4.4 Network --> Cellular

In the **Network --> Cellular** interface, Users can Enable Cellular WAN and configure Cellular.

Note: APN cannot be empty.


LoRa ▼ LoRaWAN ▼ Modbus ▼ Network ▼ System ▼ Server ▼ LogRead ▼ Home Logout

Cellular Settings

☐ Enable Cellular WAN

IMEI **866308064624893**

APN

Service UMTS / GPRS ▼

Dial Number

Pincode

Username

Password Show

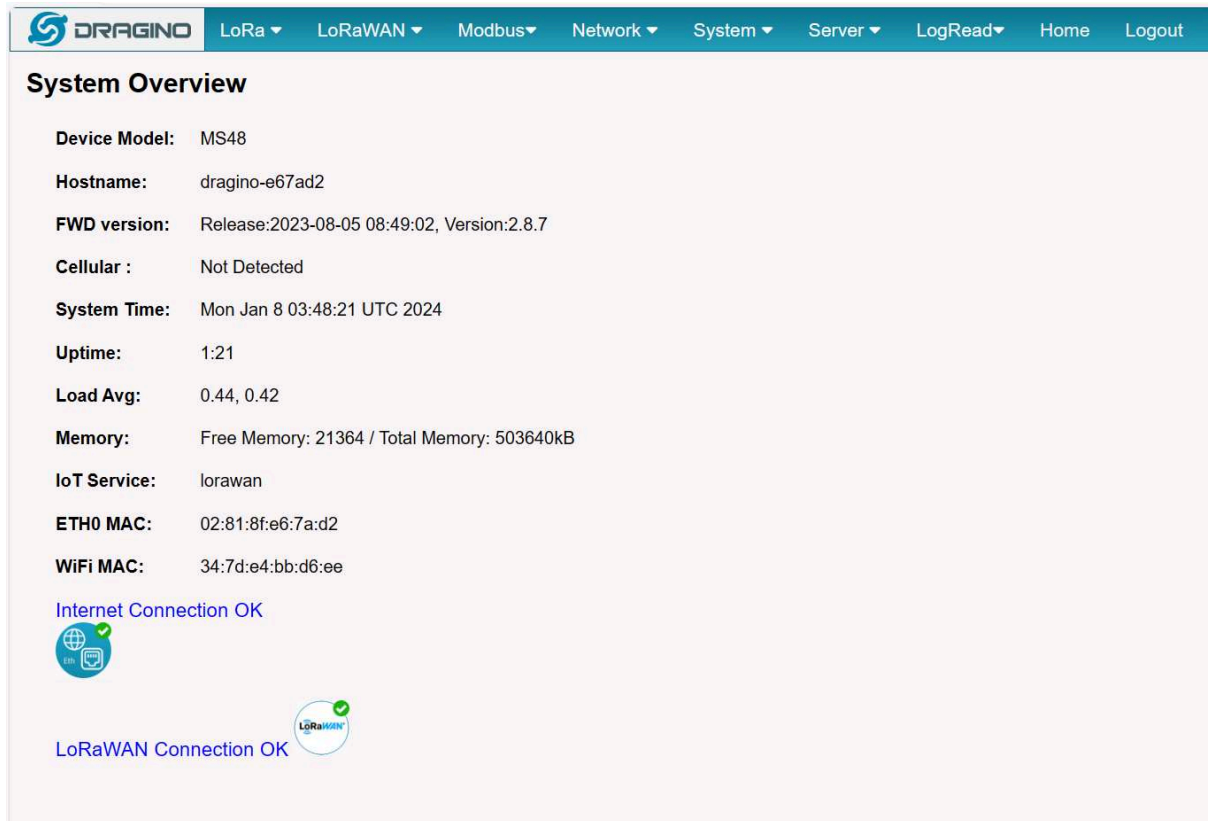
Save&Apply Cancel

After the configuration is complete, return to the Home interface and put the mouse on the Cell icon to check the Cellular state.

3.5 System

3.5.1 System --> System Overview

Shows the system info:



The screenshot shows the 'System Overview' page of a Dragino device. The top navigation bar includes links for LoRa, LoRaWAN, Modbus, Network, System, Server, LogRead, Home, and Logout. The main content area displays various system parameters:

- Device Model:** MS48
- Hostname:** dragino-e67ad2
- FWD version:** Release:2023-08-05 08:49:02, Version:2.8.7
- Cellular :** Not Detected
- System Time:** Mon Jan 8 03:48:21 UTC 2024
- Uptime:** 1:21
- Load Avg:** 0.44, 0.42
- Memory:** Free Memory: 21364 / Total Memory: 503640kB
- IoT Service:** lorawan
- ETH0 MAC:** 02:81:8f:e6:7a:d2
- WiFi MAC:** 34:7d:e4:bb:d6:ee

Below the system info, there are two status indicators:

- Internet Connection OK:** Represented by a globe icon with a green checkmark.
- LoRaWAN Connection OK:** Represented by a LoRaWAN logo icon with a green checkmark.

3.5.2 System --> System General

There are two login for MS48-LR: **root /dragino** or **admin /dragino**. Both root and admin has the same right for WEB access. But root user has also the right to access via SSH to Linux system. admin only able to access WEB interface.

This page can be used to set the password for them.

Timezone: Set device timezone.

Time Synchronization Service: Set the time synchronization server.

HTTP Web Service: Enable/Disable the HTTP service via WAN interface.

Terminal Service: Enable/Disable the SSH service via WAN interface.

Fallback Settings: Enable/Disable the Fallback interface.

[Keepalive_Script](#): Set the keepalive_script interval.

 LoRa ▼ LoRaWAN ▼ Modbus ▼ Network ▼ System ▼ Server ▼ LogRead ▼ Home Logout

System General

System Password

Password

Show

SetPassword

Login: root

Password (admin)

Show

SetAdminPassword

TimeZone

Timezone

UTC ▼

SetTimezone

Time Synchronization Service

Please enter the time synchronization address, support multiple server addresses

Set TimeSync

HTTP Web Service

Enable HTTP Service

☒

Set HTTP Port

80

Set HTTP Service

Terminal Service

Enable SSH service

☒

Set SSH Port

22

Set SSH Service

FallBack Service

Enable FallBack service

☒

Set FallBack Address

172.31.255.254

Set Fallback Service

Keep Alive

Network Check Time

15

Set Check Time

3.5.3 System --> Backup/Restore

 DRAGINO

LoRa ▾

LoRaWAN ▾

Modbus ▾

Network ▾

System ▾

Server ▾

LogRead ▾

Home

Logout

Backup/Restore

Click "Generate archive" to download a tar archive of the current configuration files."

Download backup: [Download Backup File](#)

To restore configuration files, you can upload a previously generated backup archive here.

Restore backup: 未选择文件

3.5.4 System --> Remotedit

In the **System-> Remotedit** interface, users can configure the gateway to be accessed remotely via Remote.it.
the users can refer to this link to configure them: [Monitor & Remote Access Gateway](#)

 DRAGINO

LoRa ▾

LoRaWAN ▾

Modbus ▾

Network ▾

System ▾

Server ▾

LogRead ▾

Home

Logout

Remote.it

1. Install Remote.it

2. Register

Bulk ID Code / Licence Key

3. Remove

To change registration, please Remove and Install again.

Status

Remoteit is not installed

Device is not registered

3.5.5 System --> Package Management

In the **System --> Package Management** interface, Users can check the current version of Core Packages.


LoRa ▼ LoRaWAN ▼ Modbus ▼ Network ▼ System ▼ Server ▼ LogRead ▼ Home Logout

Package Management

General Settings

Enable update every boot ☒
Enable update every day midnight ☒
SAVE

Core Packages

Name	Current Version
dragino-httpd :	2023-04-07
dragino-ui :	2023-10-30
dragino-fwd :	2023-08-18
dragino-ups :	2023-06-30
dragino-fallback :	23.01.05
armbian-bsp-cli-draginohp0z :	23.02.16
linux-image-current-draginohp0z :	22.05.2

Manual_Update

Package Auto-Update Log

4. Build-in Server

the default factory version of MS48-LR is installed with the built-in Applicant server: **Node-Red**, LoRaWAN Server: **ChirpStack**.

Note:

Path: Server --> Network Server

Server --> Application Server

Troubleshooting:

1. URL does not jump properly

For the ChirpStack, you can use the local IP address and the port is **8080** to access it.

For the Node-Red, you can use the local IP address and the port is **1880** to access it.

4.1 LoRaWAN Network Server -- ChirpStack

You can access the gateway's built-in LNS server of **ChirpStack** via the URL(<http://<hostname>:8080> or <http://<local-IPV4-address>>) in your browser.

Such as <http://dragino-54ff12:8080> or <http://<Local-IPV4-Address>>

Login account:

Username : admin

Password: admin

4.2 Application Server -- Node-Red

You can access the gateway's built-in AS server of **Node-Red** via the URL(<http://<hostname>:1880> or <http://<local-IPV4-address>>) in your browser.

Such as <http://dragino-54ff12:1880> or <http://<Local-IPV4-Address>>

Using Node-Red, InfluxDB and Grafana

The MS48-LR supports this combination, the default, Node-red is pre-installed but the InfluxDB and Grafana is not pre-installed.

the users can refer to this link to install them.

<http://wiki.dragino.com/xwiki/bin/view/Main/Armbian%20OS%20instruction/#H2.6HowtoinstallGrafanaandinfluxdb>

Upgrade the node.js

By default, the MS48-LR node.js uses the pre-install version v12 which is due to Debian the ultra-stable via ultra-old.

the users can refer to this link to upgrade them.

<http://wiki.dragino.com/xwiki/bin/view/Main/Armbian%20OS%20instruction/#H2.5Howtoupgradethenodejsversiontothelatest>.

5. Use RS232 Interface

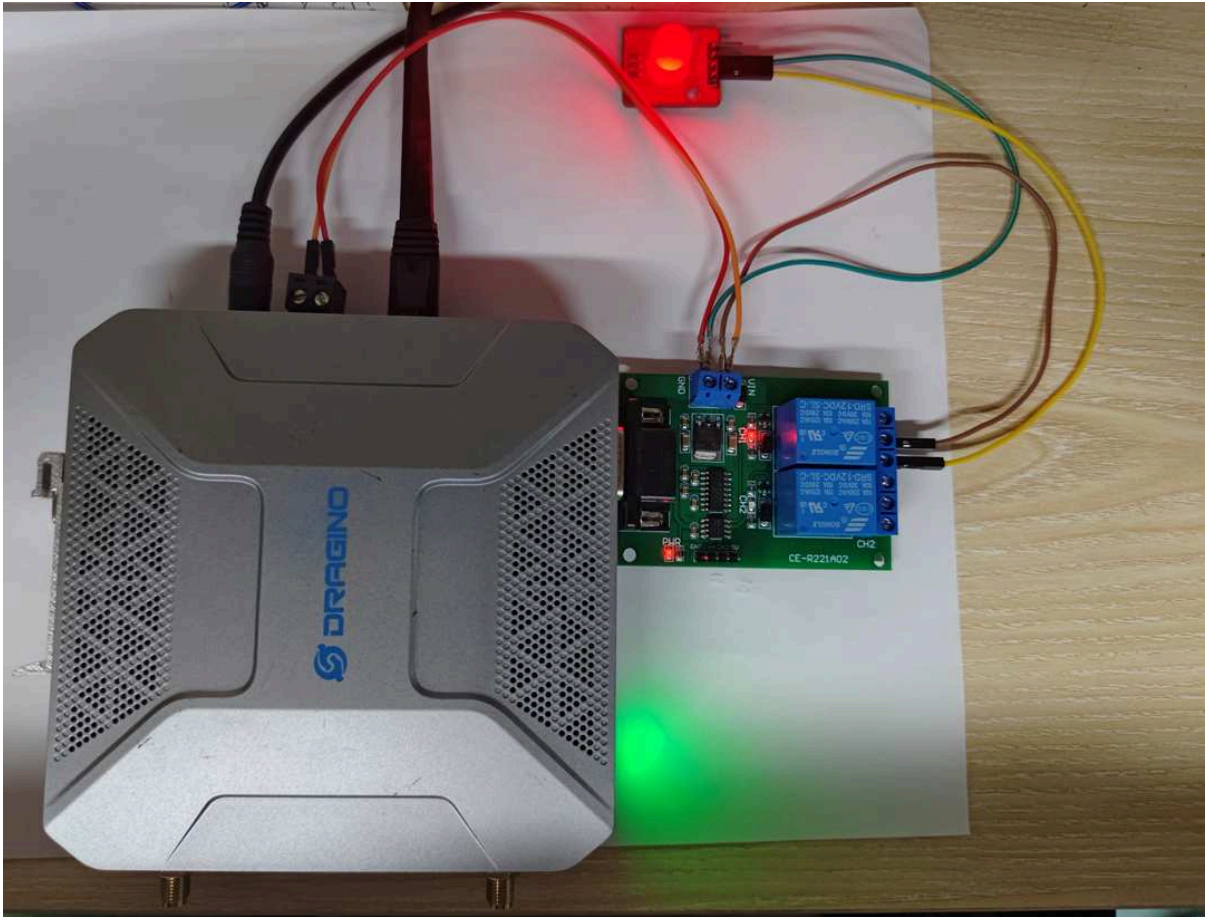
MS48-LR includes a local ChirpStack Server and Node-Red. This example shows how to configure LHT65N to use with the local Node-Red server. This example assumes users already have:

- LHT65N register on MS48-LR Built-In ChirpStack server already
- The user is able to see the data on the built-in ChirpStack server device page
- The RS232 relay is connected to the RS232 interface of the MS48-LR

The MS48-LR RS232 interface corresponds to [/dev/ttyS2](#).

Below are the steps for the MS48-LR read LHT65N's temperature control RS232 relay example:

Example of RS232 relay control via LoRaWAN:



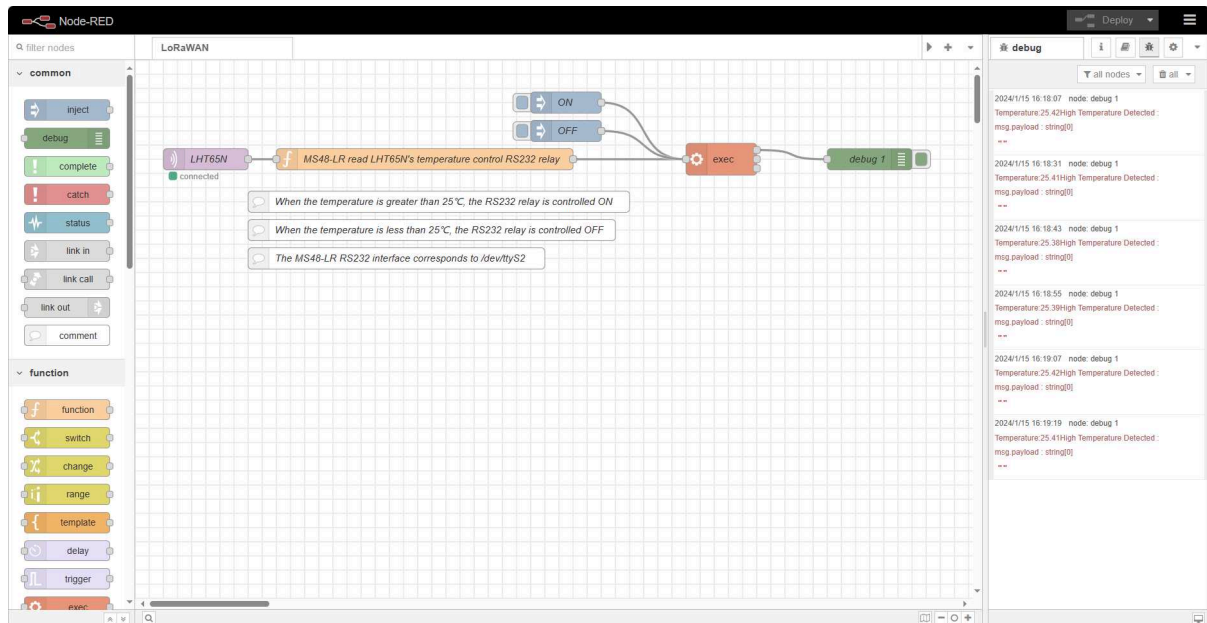
5.1 Link Node-Red to Local ChirpStack

Users can download the Node-Red decoder from this link and import it into the Node-Red platform: [MS48-LR read LHT65N's temperature control RS232 relay.json](#)

For more information on importing Input Flow, check out this link: [Import Input Flow for Dragino Sensors](#)

After importing the Input Flow is complete, the user needs to edit the MQTT in the node

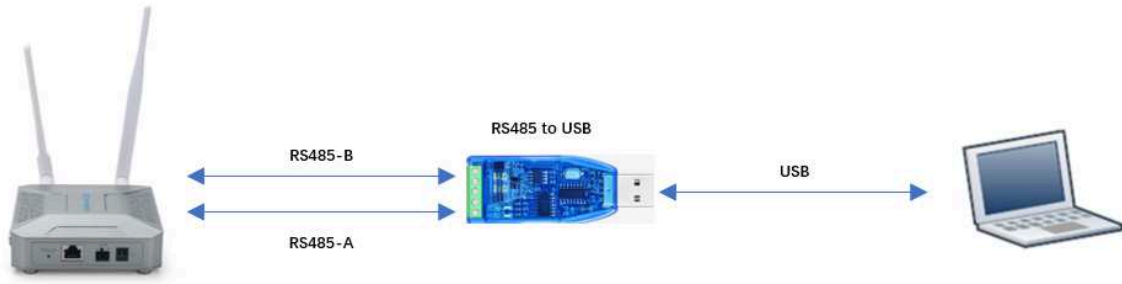
The specific steps can be found at this link: [Example: Use Local Server ChirpStack and Node-Red](#)



6. Use RS485 Interface

The MS48-LR RS485 interface corresponds to [/dev/ttyS1](#).

Below are the steps for the MS48-LR RS485 interface to manually send and receive data:



6.1 Initialize the GPIO21

Users need to run the following command to configure GPIO21:

```
echo 21 > /sys/class/gpio/export
echo "out" > /sys/class/gpio/gpio21/direction
```

6.2 Set the RS485 Tx Mode

Set the MS48-LR RS485 port to Tx mode by lowering the GPIO21 level:

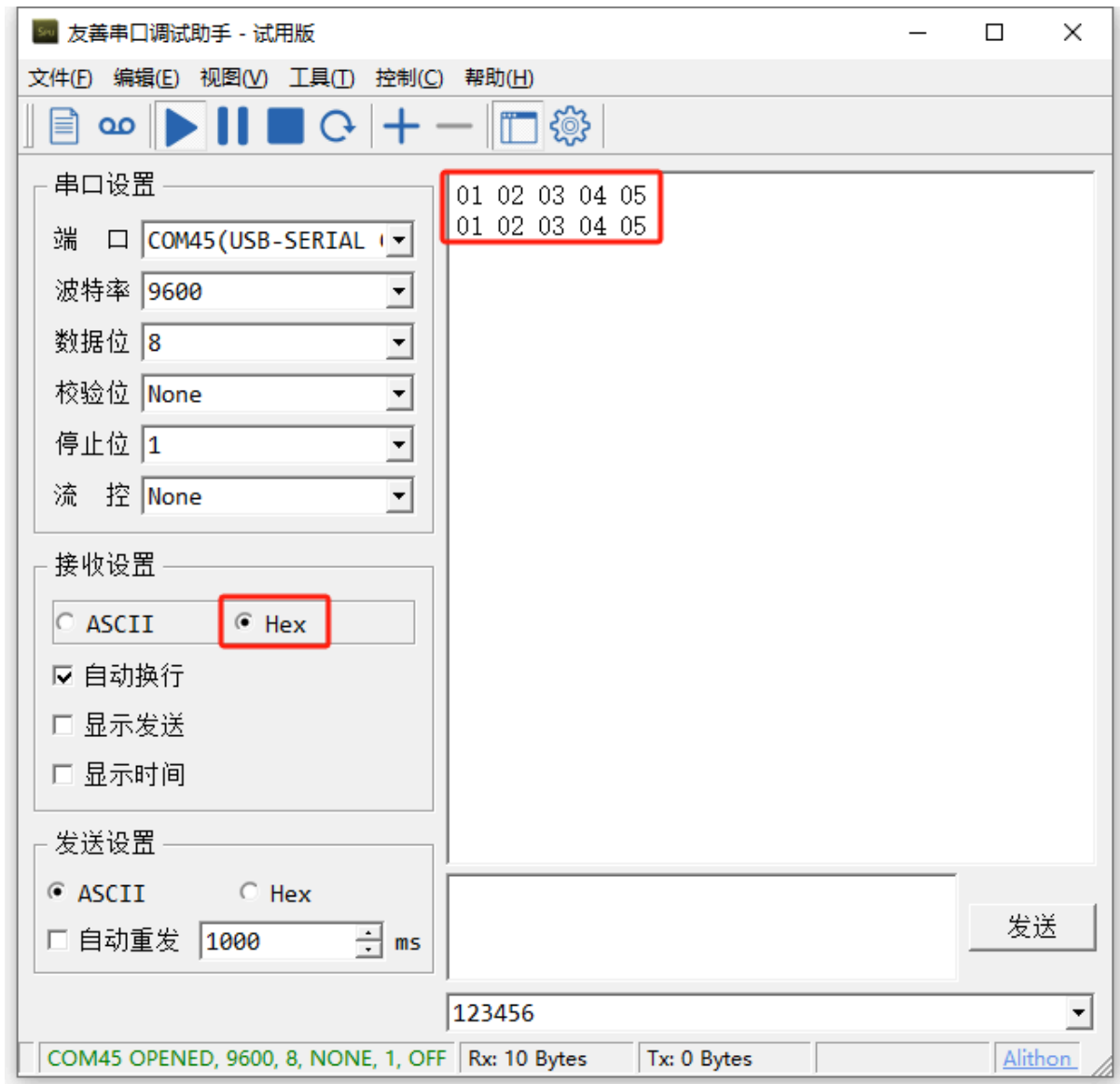
```
echo "0" > /sys/class/gpio/gpio21/value
```

Run the following command to send hexadecimal data:

```
echo -en "\x01\x02\x03\x04\x05" > /dev/ttyS1
```

```
root@dragino-2d5d26:~# echo "0" > /sys/class/gpio/gpio21/value
root@dragino-2d5d26:~# echo -en "\x01\x02\x03\x04\x05" > /dev/ttyS1
root@dragino-2d5d26:~#
root@dragino-2d5d26:~#
root@dragino-2d5d26:~#
```

Users can use the serial port tool to check the data sent by MS48-LR RS485:



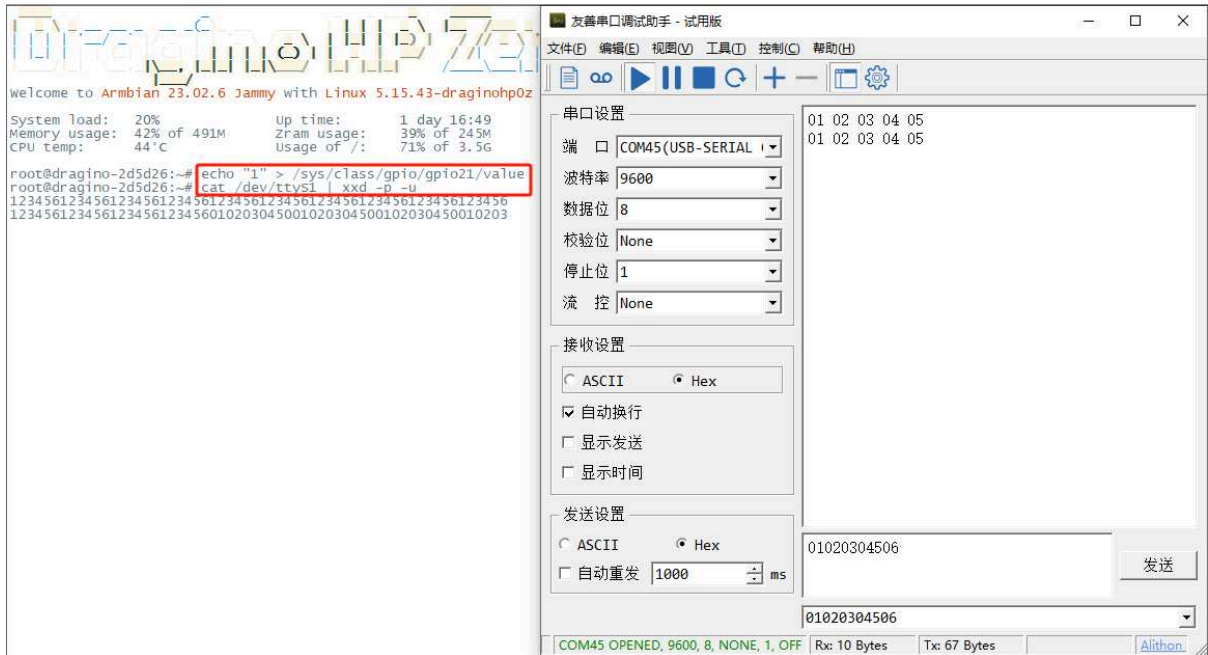
6.3 Set the RS485 Rx Mode

Set the MS48-LR RS485 port to Rx mode by pulling up the GPIO21 level:

```
echo "1" > /sys/class/gpio/gpio21/value
```

Run the following command to check the data received by the MS48-LR RS485:

```
cat /dev/ttyS1 | xxd -p -u
```



7. Power Analyze

Users can refer to and calculate the power consumption of the MS48-LR in this:

Model	Input Power	Test Mode	Current(mA)	Power Consumption(W)
MS48-LR	(+12V)	Starting current	90-190	1.080-2.280
		No network	179-210	2.148-2.520
		wifi and internet connection(LORA Start)	137-178	1.644-2.136
		wifi and internet connection(LORA Start) and 4G	144-201	1.728-2.412
		wifi and internet connection(LORA Start) and 4G and 64g U disk	172-225	2.100-2.700
		4G connection(LORA Start)	161-226	1.932-2.712
		wifi and internet connection(LORA Start) and 64g U disk	162-193	1.956-2.316
		send manually	377	4.524
		accept manually	175	2.100

8. More Services